Group Assignment # 1
Instrumentation & Measurement

Faculty of Electrical Engineering
Universiti Teknologi Malaysia

Objective:

To enhance your theoretical understanding of measurement errors, uncertainty analysis, and analog instrumentation through literature, simulation, and real-world applications in your respective fields — without requiring lab tools or physical instruments.

Assignment Scope:

Each group will select one (1) of the following interdisciplinary topics to explore. You must incorporate the following:

- Explanation of measurement errors and uncertainty based on your selected application.
- Discussion on analog instrument principles and usage.
- Application of error analysis or measurement modeling (simulation or conceptual).
- Suggestion of improvements to measurement reliability in your field.

Topic Options (Choose One):

(1) Biomedical Engineering

- Uncertainty and Error Analysis in Non-Invasive Blood Pressure Measurement
- Explain the types of errors (gross, systematic, random) that might arise using analog sphygmomanometers.
- Describe how PMMC voltmeters are related to biomedical analog signals.
- Evaluate uncertainty propagation in systolic/diastolic readings (conceptually).
- Use numerical examples to show error deviation if calibration drift occurs.

(2) Chemical & Process Engineering

- Analog Instrumentation in Monitoring Temperature & Pressure in Reactors
- Discuss the measurement uncertainty when using thermocouples or analog pressure gauges.
- Simulate a situation where reading error affects the pressure-time graph.
- Explain how analog voltmeters or ammeters contribute to system monitoring.
- Create a conceptual block diagram showing analog data acquisition.

(3) Artificial Intelligence and IoT Systems

- Impact of Sensor Error and Analog-to-Digital Conversion on AI-Based Predictions
- Describe the role of analog sensors in collecting training data.
- Explain error propagation due to ADC resolution (using formulas).
- Discuss loading effect and signal conditioning for digital conversion.
- Compare 8-bit vs 12-bit ADCs in terms of step size and voltage error.

(4) Social Sciences / Psychology

- Survey Bias and Analog Behavior Tracking in Human Response Systems
- Explore error sources in manually recorded data (e.g., stopwatches, gauges).
- Use standard deviation and uncertainty analysis in survey scores.
- Include a case study on visual analog scale (VAS).
- Simulate data rejection using Chauvenet's criterion.

Report Format:

Section	Description
Cover Page	Title, Group members, Program
Introduction	Background and importance of measurement
Theoretical Framework	Explanation of analog instruments and error types
Application Analysis	Apply to selected field; include uncertainty calculation examples
Data Simulation/Example	Provide calculations or concept simulations (no physical tools)
Improvement Recommendations	How to minimize errors or improve accuracy
Conclusion	Summary of findings
References	Minimum 3 references (APA style)

Report Guidelines:

- Length: 3–5 pages (excluding references)
- Format: Times New Roman, 12 pt, 1.5 spacing
- Submission: PDF format

Assessment Rubric (20%):

Criterion	Marks
Accuracy of Theory and Concepts	5
Relevance to Selected Field	5
Clarity of Error/Uncertainty Analysis	5
Structure and Presentation	5